

X-ray Diffraction in Single Crystals: Bragg and Laue Methods

J. W. Bowley

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X-ray diffraction was used to investigate the crystal structure of single-crystal sodium chloride using the Bragg method, with complementary Laue diffraction measurements used to assess crystal orientation and symmetry. Bragg scans were performed for NaCl crystals with (100), (110) and (111) orientations, and dominant allowed reflections were identified in each case. From fully calibrated measurements, a lattice parameter of $a = (5.61 \pm 0.03) \times 10^{-10}$, m was obtained, in good agreement with literature values. Using this lattice parameter and the face-centred cubic structure of NaCl, the crystal density was calculated to be $\rho = (2.18 \pm 0.04) \times 10^3$, kg, m⁻³. Angular calibration of the goniometer was found to dominate the experimental uncertainty. Transmission Laue diffraction of a LiF (100) crystal produced a pattern with four-fold rotational symmetry, confirming the expected crystal orientation and supporting the assumptions used in the Bragg analysis.

1 Introduction

The early 20th century was a period of great discovery in the field of X-rays and crystallography. In 1905, Albert Einstein extended Planck's theory of light being quantised by showing that the photoelectric effect means that light can behave like particles [1] as well as like waves [2]. While there existed evidence for both particle and wave behaviour of light, in 1912 this was still being debated. Similarly, it was theorised that crystals are made up from atoms arranged in regularly ordered, repeating geometric patterns in three dimensions, but no one had yet proved that crystals have this structure [3].

In February 1912, Laue theorised that the much shorter electromagnetic rays, which X-rays were supposed to be, would cause diffraction or interference phenomena in a crystal medium [4]. In April of this same year [3], he showed that X-rays are diffracted by crystal lattices and form diffraction patterns on photographic film. He proved in a single experiment that X-rays are wave-like in nature, and that crystals have a lattice-like structure [5].

Despite this breakthrough, it wasn't clear yet if crystal structure affected the X-ray diffraction pattern. In 1914, William Lawrence Bragg proposed a mathematical relationship between

atomic plane separation, X-ray wavelength and diffraction angle. At the same time, his father William Henry Bragg invented the X-ray spectrometer. Together, they proved the mathematical relationship known as Bragg's Law [5].

While Laue diffraction is not derived from Bragg's Law historically, it is now understood that both emerge from the same diffraction condition [6]. Modern scientists have adapted Bragg's and Laue's discoveries into methods of identifying orientation and physical characteristics of large crystal structures.

Both the Bragg and Laue methods use the constructive interference of X-rays when scattered by periodic atomic planes in a crystal to identify key aspects of their orientation and geometry. However, they differ on what is fixed and what is varied.

In the Bragg method, the incident X-ray is monochromatic, with a known wavelength characteristic of the emission anode. Both the crystal and detector are rotated relative to the X-ray beam, and diffraction is observed when the Bragg condition [6] is satisfied. To determine the angle that satisfies the Bragg condition, the method involves scanning the scattering angle and recording scattered X-ray intensity as a function of angle. The measured diffraction angles are then used to infer interplanar spacing and lattice parameters.

There are two main categories of Laue diffraction: back reflection and transmission [7]. This report will refer to transmission Laue, as the crystal samples that were available for this experiment were thin.

For transmission Laue, a polychromatic ('white') X-ray beam is incident on a fixed crystal. Many wavelengths are present simultaneously. For each lattice plane, there is a wavelength that satisfies the Bragg condition without rotating the crystal. Diffraction occurs in many directions, causing a pattern of spots on an X-ray detector screen. The positions of these spots encode information about the orientation, symmetry and allowed lattice planes of the crystal. Unlike the Bragg method, X-ray wavelengths are not known and must be inferred.

The main distinction is that the Bragg method varies the geometry, while the wavelength is known, whereas in the Laue method, the geometry is fixed and the wavelength ranges. The Bragg method is useful for finding the lattice spacings of crystals [8], while the Laue method is good at determining a large crystal's orientation and quality [9].

This report primarily aims to use the Bragg method to quantitatively determine the lattice parameters of NaCl (100), (110) and (111) crystals. These lattice parameters are then used to calculate the density of NaCl and compare it with standard literature values. The secondary aim is to qualitatively confirm the crystallographic orientation and symmetry of a LiF (100) crystal using the Laue method.

2 Theory

2.1 Bragg Diffraction

William Lawrence Bragg's approach to deriving the mathematical relationship between atomic plane separation, X-ray wavelength and diffraction angle is a Huygens Construction for a reflected wave [10]. This relationship is known as Bragg's Law:

$$n\lambda = 2d \sin \theta \quad (1)$$

where n is the order of diffraction, λ is X-ray wavelength, d is spacing between adjacent planes (see Equation 2), and θ is then angle between incident X-ray beam and the set of crystal planes. Refer to pages 3 to 5 inclusive in accompanying lab book for full Bragg's Law derivation.

In Bragg diffraction, when scanning intensity as a function of angle the angle with peak intensity satisfies Bragg's Law (Equation 1) and gives the spacing between adjacent planes d (Equation 2).

2.2 Miller Indices and Crystal Structure

Miller Indices (hkl) are a set of numbers which quantify the intercepts of where a crystal plane intersects the main crystallography axis of a solid [11]. They can be used to uniquely identify the plane or surface.

The spacing between adjacent planes d in the set having Miller Indices (hkl) is:

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad (2)$$

This diagram shows the crystal structure of the material studied in this experiment:

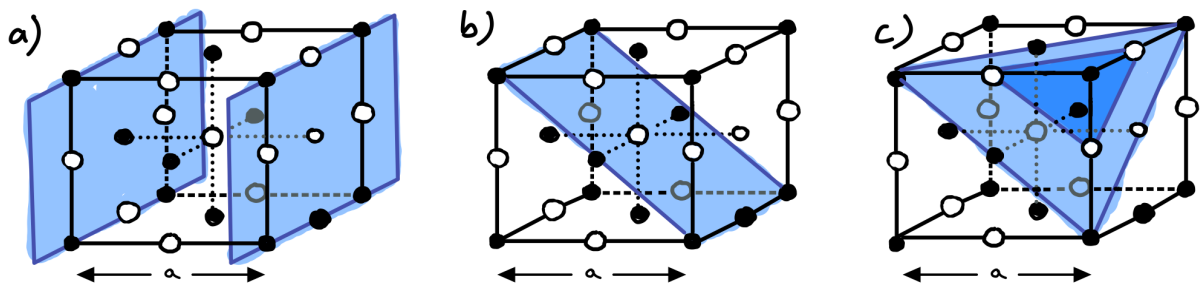


Figure 1: Crystal structure of NaCl with Miller Indices a) 100, b) 110, c) 111. Consists of the face-centred cubic (FCC) lattice [12], with a basis of two atoms: Na (filled circles) at (0,0,0) and Cl (open circles) at (0.5,0.5,0.5). Figure adapted from the laboratory manual [13].

For each Bravais lattice, selection rules determine which lattice planes produce coherent X-ray scattering [14]:

Table 1: Selection rules for coherent scattering in cubic Bravais lattices. Here n is an integer, and expressions involving n refer to even/odd.

Lattice type	Allowed reflections (hkl)	Forbidden reflections (hkl)
SC	All (hkl)	None
BCC	$h + k + l = 2n$	$h + k + l = 2n + 1$
FCC	h, k, l all even or all odd	Mixed even and odd

As NaCl is an FCC [12], then only hkl all even, or all odd, will produce coherent scattering.

2.3 Laue Diffraction Geometry

In the Laue method, a collimated X-ray beam containing a continuous range of wavelengths is incident on a stationary single crystal. The beam is directed along a crystallographic direction of high symmetry, typically normal to a low index plane (such as 100) as shown schematically:

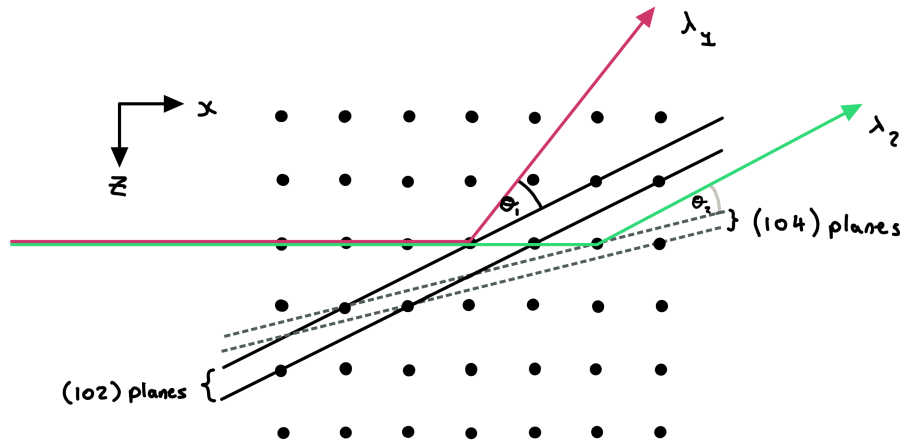


Figure 2: Schematic diagram of Laue Geometry, adapted from the laboratory manual [15]

For each set of lattice planes, the direction of constructive scattering is fixed by geometry. Because the incident beam is polychromatic ('white'), there exists a wavelength that satisfies the Bragg condition (Equation 1) for each plane without rotating the crystal. As a result, multiple reflections occur simultaneously for different planes, producing an array of diffraction spots.

The direction of each diffracted beam depends on the orientation of the reflecting lattice planes:

The symmetry of the Laue diffraction pattern reflects the symmetry of the crystal about the incident beam direction. Thus the spot positions encode orientation/symmetry of the large crystal structure. In this experiment, Laue diffraction was used to assess crystal orientation and symmetry rather than to perform a full quantitative indexing of diffraction spots.

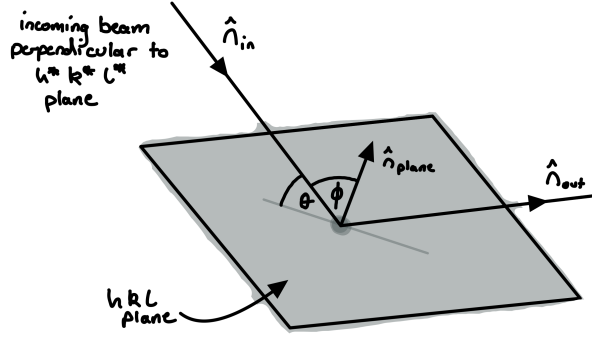


Figure 3: Vector geometry of reflection from general $((hkl))$ plane, adapted from the laboratory manual [15]

3 Methods

3.1 Bragg Method

A collimated beam of X-rays impinges on a single crystal mounted on a rotating stage. This stage is tilted at an angle θ to the beam, and reflected X-rays are detected by a Geiger-Müller tube at angle 2θ :

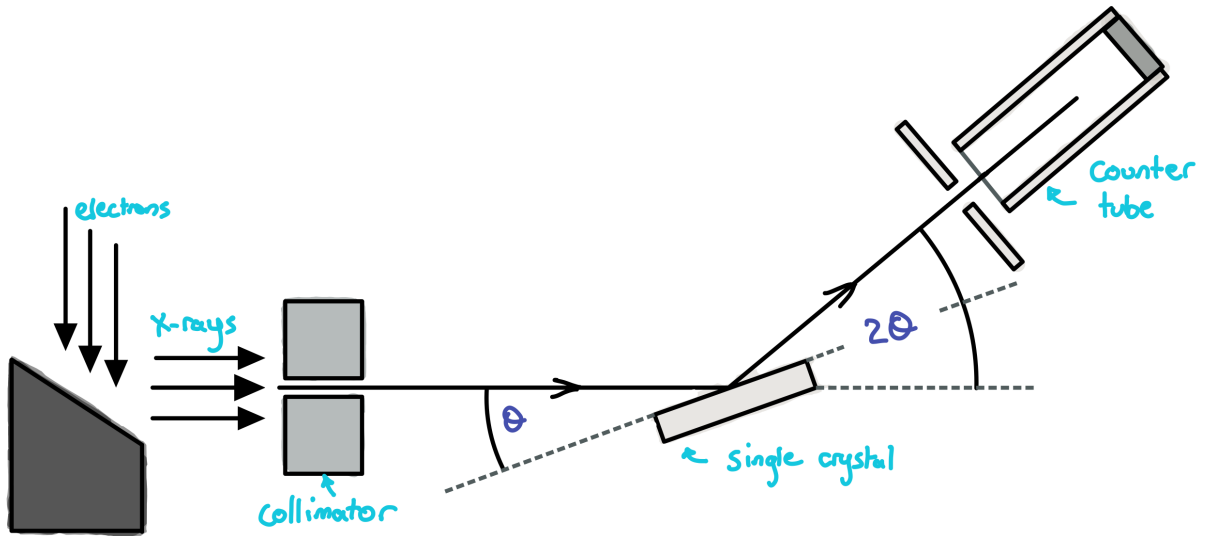


Figure 4: Schematic Diagram of Bragg experimental setup within X-ray unit, adapted from the laboratory manual [13]

The crystal and detector are rotated while the incident beam direction and wavelength remain fixed. Geiger-Müller tube measures X-ray intensity as a function of angle θ . The goal is to identify positions of peak intensity to infer plane distance d using Equation 1.

The zero-degree positions of both the sample and detector on the goniometer have to be calibrated before scans be carried out. This calibration depends on both the unit and the crystal used, so was repeated for every new crystal measured.

To perform calibration, mounted crystal to rotating stage and open the X-ray unit control software. Within the software, the crystal type was registered and calibration mode was selected. The unit then performed a brief sweep of angles, identifying calibration angles. These angles were checked with the physical angle of goniometer on unit to see if appeared visually correct.

Once calibrated, the X-ray unit was set to the following parameters (taken from laboratory manual [13]):

Table 2: X-ray unit parameters for Bragg method

Coupling mode	1:2
Angle step	0.1°
Gate time	2 s
Scan range	3° to 30°

Due to time constraints during scheduled lab sessions, not all Bragg measurements could be repeated individually. For additional completeness of analysis across different crystal orientations, additional data sets were obtained from a partner working on the same experimental setup under identical nominal conditions.

Only the NaCl (100) and NaCl (110) first measurements were fully calibrated to the knowledge of the author and performed personally. For the remaining data sets, the raw angle-intensities were analysed independently, with calibration uncertainty treated explicitly. Additional data sets obtained from a partner working on the same setup under the same conditions were:

Table 3: Additional data sets obtained from partner

Material	Measurement
NaCl (100)	Second measurement
NaCl (110)	Second measurement
NaCl (111)	First measurement
NaCl (111)	Second measurement
LiF (100)	Measurement
KBr (100)	Measurement

For full data tables, please refer to the accompanying lab book, pages 10 to 34.

Data was read into a JupyterLab notebook using `pandas` DataFrames for plotting, where they were plotted using `matplotlib.pyplot` for visualisation (see Figures 5–9 in Section 4). Reference lab book pages 35 to 58 for full details. The plotted figures and raw data tables were used in conjunction to identify angles of peak intensity and approximate full widths at half maximum (FWHM). FWHM were used to assess peak sharpness and symmetry.

3.2 Bragg Data Analysis

Raw Bragg diffraction data were recorded as count values at discrete crystal angles ω_c , using a fixed gate time of $T = 2$ s and an angular step size of 0.1° . All raw data tables are provided in the accompanying lab book (pp. 10–34). The recorded counts were treated as Poisson-distributed, with statistical uncertainty $\delta N = \sqrt{N}$, as discussed in the laboratory book (pp. 59–61).

Count rates were obtained via

$$I = \frac{N}{T}, \quad (3)$$

with corresponding uncertainty

$$\delta I = \frac{\sqrt{N}}{T}. \quad (4)$$

The instrumental resolution of the goniometer dominates angular uncertainty. Since the crystal angle was scanned in steps of 0.1° , a uniform angular uncertainty of $\delta\theta = \pm 0.1^\circ$ was assigned to all Bragg peak positions. In all cases, the estimated statistical uncertainty in peak centre location was smaller than this step size, so instrumental resolution dominated.

Bragg peak positions were identified by visual inspection of intensity versus angle plots generated in a JupyterLab notebook. Peaks were selected based on being sharp, repeatable, and symmetric across repeated scans. Full widths at half maximum (FWHM) were estimated directly from the raw data tables and used qualitatively to assess peak sharpness and symmetry rather than as an input to uncertainty propagation.

For a subset of Bragg scans acquired by a laboratory partner, calibration angles were not recorded. These scans, particularly the NaCl (111) data, exhibited a systematic angular offset consistent with incomplete or omitted zero-angle calibration. This effect was inferred from the relative displacement of expected Bragg peak positions when compared to calibrated NaCl (100) and NaCl (110) datasets. This systematic offset was treated as part of the overall angular uncertainty and does not affect peak identification or relative comparisons.

3.3 Laue Method

Laue diffraction measurements were performed using a transmission geometry with a fixed single crystal and a polychromatic X-ray beam. A LiF crystal with nominal (100) orientation was mounted such that the incident beam was approximately normal to the crystal surface. Unlike the Bragg method, the crystal and detector geometry remained fixed throughout the measurement.

The transmitted Laue pattern was recorded on a flat detector positioned downstream of the crystal. Due to the use of a white X-ray beam, multiple wavelengths were present simultaneously, allowing many sets of lattice planes to satisfy the Bragg condition without rotating the

crystal. As a result, diffraction occurred in multiple directions, producing a pattern of discrete spots.

Raw Laue images were exported and analysed using a JupyterLab notebook developed for this experiment. Image processing steps were intentionally kept minimal and consisted of background subtraction to remove large-scale intensity gradients, followed by contrast enhancement to improve visibility of diffraction spots. No geometric calibration or wavelength assignment was performed, as the aim of the Laue analysis was qualitative rather than quantitative.

The processed images were used to assess the symmetry of the diffraction pattern and to confirm the crystallographic orientation of the LiF sample. Full details of the image processing steps and intermediate outputs are provided in the accompanying lab book.

4 Results

4.1 Bragg Diffraction Results

Representative Bragg scans for each crystal orientation are shown in Figures 5–9. In all cases, dominant reflections were identified as the strongest, sharpest peaks appearing consistently across repeated scans. Weaker features at lower angles were attributed to background effects or secondary scattering and were not used in quantitative analysis.

Observed Bragg Peaks

For NaCl (100), a dominant reflection corresponding to the allowed (200) plane was observed at $\theta \approx 7.3^\circ$ (see corresponding Bragg scans in this section). The peak was sharp and symmetric, with an estimated full width at half maximum (FWHM) of approximately 0.4° , and showed good agreement between repeated measurements.

For NaCl (110), a strong and well-defined (220) reflection was observed at $\theta \approx 10.3^\circ$ (see corresponding Bragg scan in this section). This peak was particularly sharp, with an estimated FWHM of approximately 0.3° , and exhibited excellent repeatability.

For NaCl (111), the dominant (222) reflection was observed at $\theta \approx 12.6^\circ$ (see corresponding Bragg scan in this section). Both scans exhibited broader peak shapes and a systematic angular displacement relative to the calibrated NaCl (100) and NaCl (110) datasets. As discussed in the lab book, this behaviour is consistent with incomplete or omitted zero-angle calibration for these scans. Despite this, peak identification remained unambiguous.

Bragg scans were also obtained for LiF (100) and KBr (100) (see corresponding Bragg scans in this section).

In both cases, dominant reflections were visible, though with increased background noise and

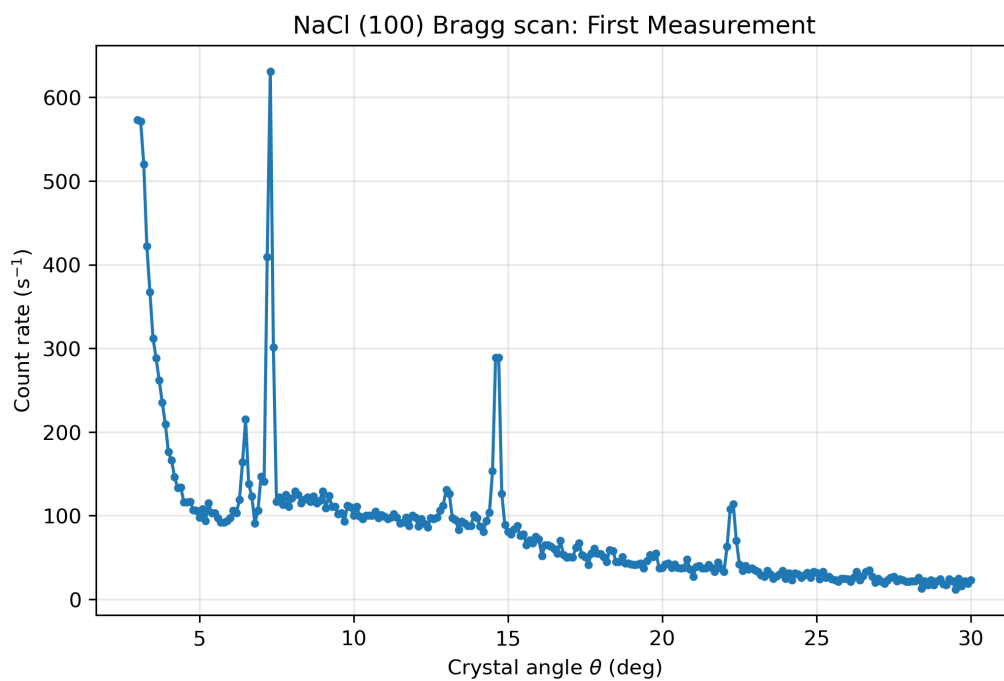


Figure 5: Bragg scan for NaCl (100), first measurement.

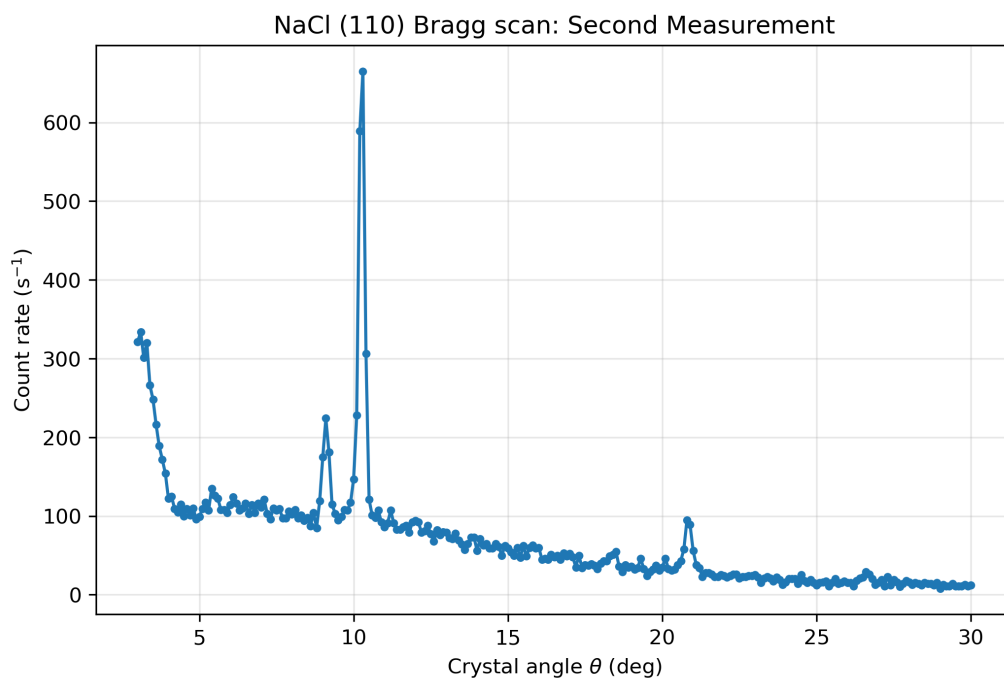


Figure 6: Bragg scan for NaCl (110), first measurement.

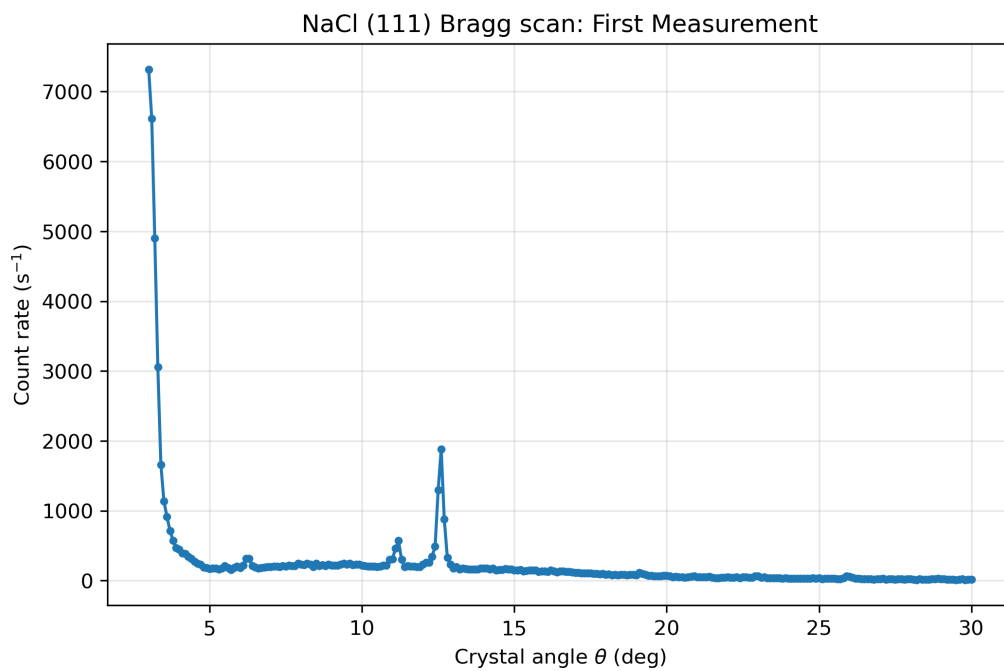


Figure 7: Bragg scan for NaCl (111), data from partner.

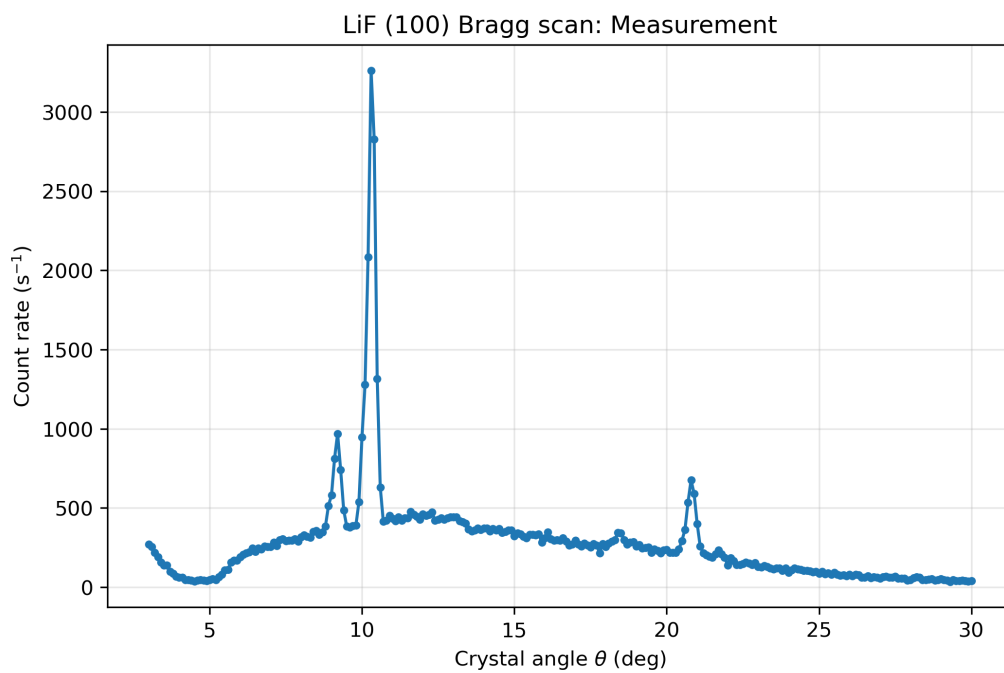


Figure 8: Bragg scan for LiF (100), data from partner.

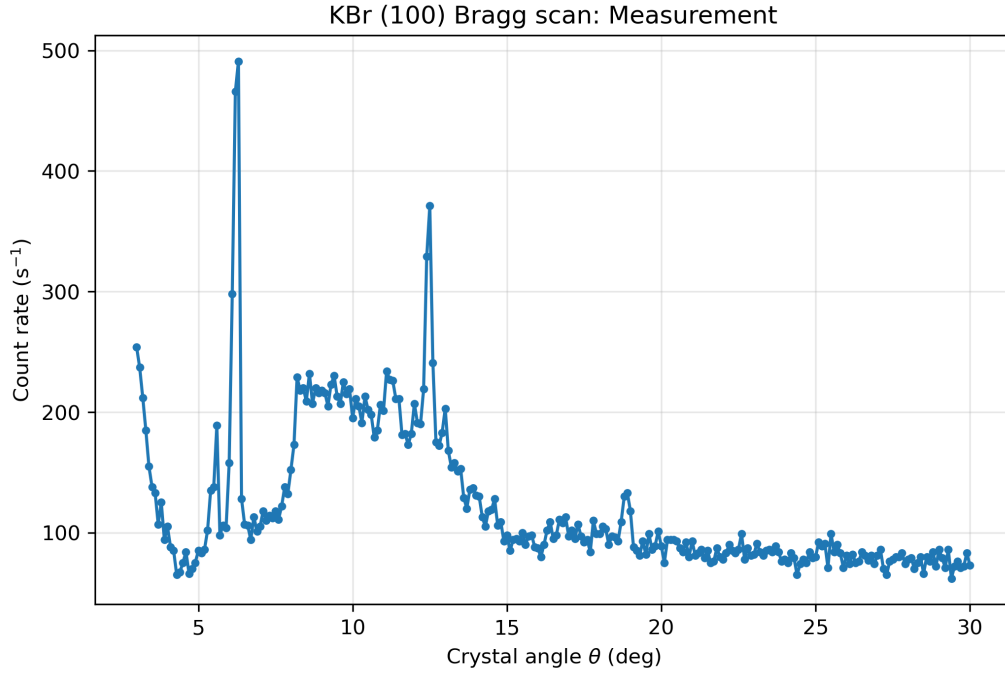


Figure 9: Bragg scan for KBr (100), data from partner.

reduced peak sharpness compared to NaCl. These datasets were therefore used primarily for qualitative comparison.

Lattice Parameter Determination

Interplanar spacings were calculated from the identified peak positions using Bragg's law for first-order reflections ($n = 1$). Lattice parameters were then inferred assuming cubic symmetry and Miller indices assigned according to FCC selection rules.

Two sets of lattice parameters and density values were calculated:

Full dataset (all Bragg scans). Using all available Bragg data, including scans acquired by a laboratory partner, the lattice parameter and density were determined to be

$$a = (5.61 \pm 0.017) \times 10^{-10} \text{ m},$$

$$\rho = (2.15 \pm 0.02) \times 10^3 \text{ kg m}^{-3}.$$

Calibrated dataset (own measurements only). Restricting the analysis to the fully calibrated NaCl (100) and NaCl (110) scans personally acquired yielded

$$a = (5.61 \pm 0.03) \times 10^{-10} \text{ m},$$

$$\rho = (2.18 \pm 0.04) \times 10^3 \text{ kg m}^{-3}.$$

Justification for Dataset Selection

As documented in the lab book, only the NaCl (100) and NaCl (110) first measurements were fully calibrated with recorded zero-angle offsets. Several additional datasets, most notably NaCl (111), exhibit a systematic angular offset consistent with incomplete calibration. While this offset does not affect peak identification or relative comparisons, it introduces a systematic uncertainty in absolute peak positions.

For this reason, results derived from the fully calibrated dataset are considered more reliable for quantitative determination of lattice parameters and density. The full dataset results are nevertheless reported for completeness and demonstrate consistency within uncertainty.

Overall, the calibrated results show good agreement with literature values for NaCl [16], confirming the validity of the Bragg method when instrumental calibration is carefully controlled.

4.2 Laue Diffraction Results

Figure 10 shows the processed Laue diffraction pattern obtained for the LiF crystal. The pattern consists of a regular array of well-defined diffraction spots arranged with clear four-fold rotational symmetry about the central transmitted beam.

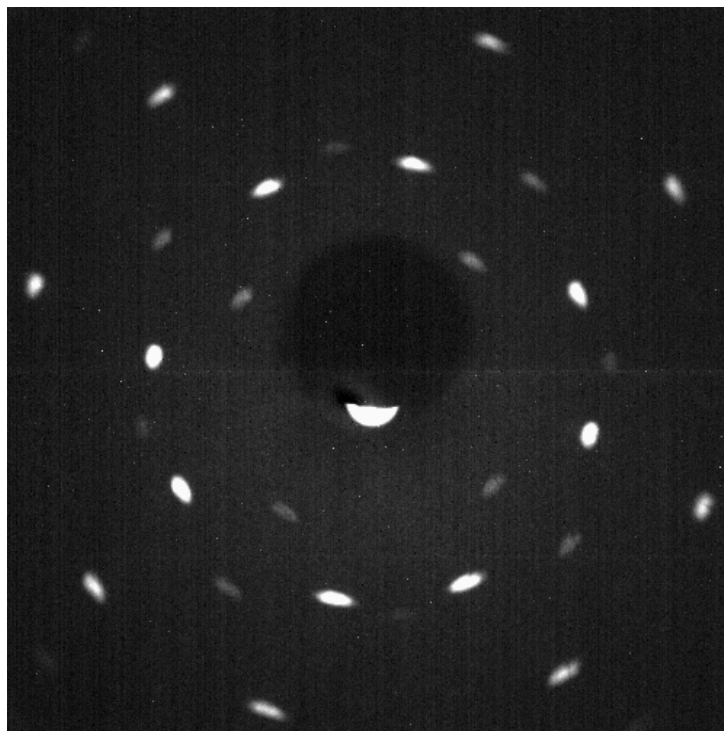


Figure 10: Processed transmission Laue diffraction pattern for the LiF (100) crystal. Contrast enhancement and background subtraction have been applied to improve spot visibility.

The observed symmetry of the diffraction pattern is consistent with a crystal oriented along a

$\langle 100 \rangle$ direction, as expected for a cubic lattice viewed normal to a (100) face. The approximately square arrangement of diffraction spots and the presence of perpendicular mirror symmetries support this assignment.

No attempt was made to assign Miller indices to individual diffraction spots or to extract lattice parameters from the Laue pattern. Instead, the Laue measurement was used to confirm crystal orientation and symmetry, providing an independent qualitative check on the assumptions used in the Bragg analysis.

5 Discussion

5.1 Bragg Method

The Bragg method provided clear and repeatable diffraction peaks for all NaCl crystal orientations studied. The strongest agreement with expected behaviour was observed for the NaCl (100) and NaCl (110) crystals, where calibration was performed explicitly, and peak positions were well-defined. The inferred lattice parameters from these orientations were internally consistent and in agreement with literature values within uncertainty.

The NaCl (111) data exhibited a systematic angular offset consistent with incomplete or omitted zero-angle calibration. While this limited the precision with which absolute lattice parameters could be extracted from these scans, it did not affect the identification of the dominant allowed reflection or relative comparison with other datasets. Treating this effect as an additional contribution to angular uncertainty is therefore justified.

Uncertainty in the inferred lattice spacing is dominated by angular uncertainty rather than counting statistics. This is supported by the fact that the estimated peak centre uncertainty from count rate fluctuations was smaller than the instrumental step size in all cases. As a result, a uniform angular uncertainty was used throughout, ensuring consistency across datasets.

Higher-order reflections were observed in some scans but were not used for quantitative analysis due to lower intensity and increased susceptibility to peak broadening and background effects. Restricting analysis to first-order dominant reflections provided the most reliable determination of lattice parameters.

Overall, the Bragg method proved effective for determining lattice spacing and validating crystal structure selection rules, particularly when calibration was performed carefully and instrumental limitations were properly accounted for.

5.2 Laue Method

Laue diffraction measurements were performed using a transmission geometry with a fixed single crystal and a polychromatic X-ray beam. A LiF crystal with nominal (100) orientation was mounted such that the incident beam was approximately normal to the crystal surface. Unlike the Bragg method, the crystal and detector geometry remained fixed throughout the measurement.

The transmitted Laue pattern was recorded on a flat detector positioned downstream of the crystal. Due to the use of a white X-ray beam, multiple wavelengths were present simultaneously, allowing many sets of lattice planes to satisfy the Bragg condition without rotating the crystal. As a result, diffraction occurred in multiple directions, producing a pattern of discrete spots.

Raw Laue diffraction images were loaded using the ImageIO Python library [17]. ImageIO was chosen because it provides reliable support for scientific image formats, including PNG and TIFF, while preserving original pixel intensities and bit depth without implicit rescaling. This ensured that subsequent image processing steps operated on data that accurately represented the detector output.

Processed images were visualised using `matplotlib.pyplot.imshow` for inspection and presentation [18]. Background subtraction and contrast enhancement were applied to the images using routines from the `scikit-image` library [19] in order to improve the visibility of diffraction spots and suppress large-scale intensity gradients. No geometric calibration or wavelength assignment was performed, as the aim of the Laue analysis was qualitative rather than quantitative.

The processed Laue images were used to assess the symmetry of the diffraction pattern and to confirm the crystallographic orientation of the LiF sample. Full details of the image processing steps and intermediate outputs are provided in the accompanying lab book.

5.3 Comparison of Methods

The Bragg and Laue diffraction methods play complementary roles in the analysis of single crystals. In this experiment, Bragg diffraction was used as the primary quantitative tool, enabling precise determination of interplanar spacings, lattice parameters, and densities through measured diffraction angles and a known X-ray wavelength. The accuracy of the Bragg method is therefore strongly dependent on angular calibration and instrumental resolution, which were identified as the dominant sources of uncertainty.

In contrast, the Laue method was used qualitatively to assess crystal orientation and symmetry. By employing a fixed crystal geometry and a polychromatic X-ray beam, Laue diffraction

allows many lattice planes to diffract simultaneously, producing a characteristic spot pattern whose symmetry reflects the underlying crystal structure. This makes the Laue method particularly effective for confirming crystal orientation and identifying high-symmetry directions, but less suitable for quantitative lattice parameter extraction without additional calibration.

Used together, the two methods provide an internal consistency check. The Laue diffraction pattern confirmed that the LiF crystal was correctly oriented along a $\langle 100 \rangle$ direction, supporting the crystallographic assumptions used in the Bragg analysis. Meanwhile, the Bragg measurements supplied the quantitative structural information that the Laue method alone could not provide.

Overall, the experiment demonstrates how Bragg and Laue diffraction are not competing techniques, but complementary ones: Bragg diffraction yields precise numerical results, while Laue diffraction provides rapid qualitative verification of crystal orientation and symmetry.

6 Conclusions

Bragg diffraction was successfully used to determine the structural properties of single-crystal sodium chloride. Clear and repeatable diffraction peaks were obtained for the (100) and (110) crystal orientations following careful angular calibration, allowing reliable determination of interplanar spacings and lattice parameters. From these measurements, a lattice parameter of $a = (5.61 \pm 0.03) \times 10^{-10} \text{ m}$ was obtained, leading to a calculated NaCl density of $\rho = (2.18 \pm 0.04) \times 10^3 \text{ kg m}^{-3}$. Both values are consistent with accepted literature values within experimental uncertainty.

Comparison between fully calibrated measurements and additional datasets highlighted that angular calibration of the goniometer was the dominant source of systematic uncertainty in the experiment. Counting statistics were found to contribute a smaller uncertainty, indicating that improvements in angular resolution and calibration procedures would lead to the greatest gains in measurement precision.

Laue diffraction measurements provided a complementary qualitative assessment of crystal orientation and symmetry. The observed four-fold symmetry of the LiF Laue pattern confirmed a $\langle 100 \rangle$ crystal orientation, supporting the crystallographic assumptions underlying the Bragg analysis. Although no quantitative indexing of Laue diffraction spots was performed, the method proved effective as a rapid check of crystal orientation and quality.

Overall, the experiment demonstrates how Bragg diffraction can be used to quantitatively determine lattice parameters and density in crystalline solids, while Laue diffraction serves as a valuable complementary technique for confirming crystal orientation and symmetry. Possible extensions include improved angular calibration, repeated Bragg scans to reduce systematic uncertainty, and full geometric calibration of the Laue setup to enable quantitative spot indexing.

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